```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>Busbar Current Capacity & Weight Calculator</title>
<script src="https://cdn.tailwindcss.com"></script>
<style>
  body { font-family: system-ui, -apple-system, "Segoe UI", Roboto, sans-serif; }
  .num { font-variant-numeric: tabular-nums; }
</style>
</head>
<body class="bg-slate-100 text-slate-800 min-h-screen">
<main class="max-w-xl mx-auto p-4 pb-10">

  <header class="py-5">
    <h1 class="text-2xl font-bold text-slate-900">Busbar Calculator</h1>
    <p class="text-sm text-slate-500 mt-1">Current capacity, size aur weight per
meter</p>
  </header>

  <!-- Inputs -->
  <section class="bg-white rounded-xl border border-slate-200 p-4 space-y-4">
    <div>
      <label for="amps" class="block text-sm font-medium mb-1">Rated current (A)
</label>
      <input id="amps" type="number" inputmode="numeric" min="1" step="1"
placeholder="e.g. 800"
        class="w-full rounded-lg border border-slate-300 px-3 py-3 text-lg focus:outline-
none focus:ring-2 focus:ring-amber-500 focus:border-amber-500">
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      <p id="err" class="text-sm text-red-600 mt-1 hidden"></p>
    </div>

    <div>
      <span class="block text-sm font-medium mb-1">Material</span>
      <div class="grid grid-cols-2 gap-2" id="matGroup">
        <button type="button" data-mat="Cu" class="mat-btn rounded-lg border px-3 py-3 font-medium">Copper</button>
        <button type="button" data-mat="Al" class="mat-btn rounded-lg border px-3 py-3 font-medium">Aluminium</button>
      </div>
    </div>

    <div>
      <label for="type" class="block text-sm font-medium mb-1">Busbar type</label>
      <select id="type" class="w-full rounded-lg border border-slate-300 bg-white px-3 py-3 focus:outline-none focus:ring-2 focus:ring-amber-500">
        <option value="1">Single flat bar per phase</option>
        <option value="2">2 flat bars per phase</option>
        <option value="3">3 flat bars per phase</option>
      </select>
    </div>

    <button id="calc" class="w-full bg-amber-500 hover:bg-amber-600 active:bg-amber-700 text-slate-900 font-semibold rounded-lg py-3 text-base focus:outline-none focus:ring-2 focus:ring-offset-2 focus:ring-amber-500">
      Calculate
    </button>
  </section>
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  <!-- Result -->
  <section id="result" class="mt-4 hidden space-y-4" aria-live="polite">
    <div class="bg-slate-900 text-white rounded-xl p-5">
      <p class="text-sm text-slate-300">Recommended size</p>
      <p id="rSize" class="text-4xl font-bold mt-1 num"></p>
      <p id="rSub" class="text-sm text-amber-300 mt-1"></p>
    </div>

    <div class="grid grid-cols-2 gap-3">
      <div class="bg-white rounded-xl border border-slate-200 p-4">
        <p class="text-xs text-slate-500">Cross-section (per phase)</p>
        <p class="text-xl font-semibold num mt-1"><span id="rArea"></span> <span
class="text-sm font-normal text-slate-500">mm²</span></p>
      </div>
      <div class="bg-white rounded-xl border border-slate-200 p-4">
        <p class="text-xs text-slate-500">Weight per meter (per phase)</p>
        <p class="text-xl font-semibold num mt-1"><span id="rWt"></span> <span
class="text-sm font-normal text-slate-500">kg/m</span></p>
      </div>
      <div class="bg-white rounded-xl border border-slate-200 p-4">
        <p class="text-xs text-slate-500">Weight per meter (3 phase)</p>
        <p class="text-xl font-semibold num mt-1"><span id="rWt3"></span> <span
class="text-sm font-normal text-slate-500">kg/m</span></p>
      </div>
      <div class="bg-white rounded-xl border border-slate-200 p-4">
        <p class="text-xs text-slate-500">Current density</p>
        <p class="text-xl font-semibold num mt-1"><span id="rCd"></span> <span
class="text-sm font-normal text-slate-500">A/mm²</span></p>
      </div>
    </div>
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    <div class="bg-white rounded-xl border border-slate-200 p-4">
      <div class="flex justify-between text-sm mb-2">
        <span class="text-slate-600">Rated capacity of this size</span>
        <span class="font-semibold num" id="rCap"></span>
      </div>
      <div class="h-2.5 bg-slate-200 rounded-full overflow-hidden">
        <div id="bar" class="h-full rounded-full"></div>
      </div>
      <p id="rUtil" class="text-xs text-slate-500 mt-2"></p>
    </div>

    <div class="bg-white rounded-xl border border-slate-200 p-4">
      <p class="text-sm font-medium mb-2">Next available sizes</p>
      <ul id="alts" class="divide-y divide-slate-100 text-sm"></ul>
    </div>
  </section>

  <p class="text-xs text-slate-500 mt-6 leading-relaxed">
    Indicative values: bare flat bars, indoor, 35 °C ambient, ~50 K temperature rise, AC supply.
    Aluminium = 0.76 × copper rating; 2 bars = 1.8×, 3 bars = 2.4× (bars spaced by one thickness).
    Weight densities: Cu 8960, Al 2700 kg/m³. Final design ke liye IEC 61439 / manufacturer tables aur
    short-circuit withstand verify karein.
  </p>
</main>

<script>
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(function () {
  // [width mm, thickness mm, single-bar copper rating in A]
  const SIZES = [
    [15,3,160],[20,3,205],[20,5,300],[25,5,365],[30,5,430],[40,5,540],
    [50,5,665],[60,5,790],[40,10,750],[50,10,950],[60,10,1150],[80,10,1500],
    [100,10,1850],[120,10,2200],[160,10,2800],[200,10,3400]
  ];
  const MAT = { Cu: { name: "Copper", rho: 8960, f: 1 }, Al: { name: "Aluminium", rho: 2700, f: 0.76 } };
  const TYPE_F = { 1: 1, 2: 1.8, 3: 2.4 };

  let mat = "Cu";
  const $ = id => document.getElementById(id);

  function paintMat() {
    document.querySelectorAll(".mat-btn").forEach(b => {
      const on = b.dataset.mat === mat;
      b.className = "mat-btn rounded-lg border px-3 py-3 font-medium " +
        (on ? "bg-slate-900 text-white border-slate-900" : "bg-white text-slate-700 border-slate-300");
      b.setAttribute("aria-pressed", on);
    });
  }
  document.querySelectorAll(".mat-btn").forEach(b => b.addEventListener("click", () => { mat = b.dataset.mat; paintMat(); }));
  paintMat();

  function rating(s, n) { return s[2] * MAT[mat].f * TYPE_F[n]; }
  const fmt = (v, d) => v.toLocaleString("en-IN", { minimumFractionDigits: d, maximumFractionDigits: d });
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  function calculate() {
    const I = parseFloat($("amps").value);
    const n = parseInt($("type").value, 10);
    const err = $("err"), res = $("result");
    err.classList.add("hidden");
    if (!(I > 0)) {
      err.textContent = "Valid rated current (A) daalein.";
      err.classList.remove("hidden"); res.classList.add("hidden"); return;
    }
    const sorted = SIZES.map(s => ({ s, a: s[0] * s[1] * n }))
                        .sort((x, y) => rating(x.s, n) - rating(y.s, n));
    const idx = sorted.findIndex(o => rating(o.s, n) >= I);
    if (idx === -1) {
      const max = rating(sorted[sorted.length - 1].s, n);
      err.textContent = "Is type me max ~" + fmt(max, 0) + " A tak hi available hai. 2 ya 3
bars per phase choose karein ya parallel runs use karein.";
      err.classList.remove("hidden"); res.classList.add("hidden"); return;
    }
    const pick = sorted[idx], [w, t] = pick.s;
    const area = w * t * n, cap = rating(pick.s, n);
    const wt = area * MAT[mat].rho / 1e6;
    const util = I / cap * 100;

    $("rSize").textContent = (n > 1 ? n + " × " : "") + w + " × " + t + " mm";
    $("rSub").textContent = MAT[mat].name + " · " + (n === 1 ? "single bar" : n + " bars") +
" per phase · Width × Thickness";
    $("rArea").textContent = fmt(area, 0);
    $("rWt").textContent = fmt(wt, 2);
    $("rWt3").textContent = fmt(wt * 3, 2);
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    $("rCd").textContent = fmt(I / area, 2);
    $("rCap").textContent = fmt(cap, 0) + " A";
    $("bar").style.width = Math.min(util, 100) + "%";
    $("bar").className = "h-full rounded-full " + (util > 90 ? "bg-red-500" : util > 75 ? "bg-amber-500" : "bg-emerald-500");
    $("rUtil").textContent = "Loading " + fmt(util, 0) + "% of rated capacity" + (util > 90 ? " — margin kam hai, ek size upar consider karein." : ".");

    const alts = sorted.slice(idx + 1, idx + 4);
    $("alts").innerHTML = alts.length ? alts.map(o => {
      const [aw, at] = o.s;
      return '<li class="flex justify-between py-2 num"><span class="font-medium">' + (n > 1 ? n + " × " : "") + aw + " × " + at +
        ' mm</span><span class="text-slate-500">' + fmt(rating(o.s, n), 0) + " A · " + fmt(o.a * MAT[mat].rho / 1e6, 2) + " kg/m</span></li>";
    }).join("") : '<li class="py-2 text-slate-500">Isse bada size is table me nahi hai.</li>';

    res.classList.remove("hidden");
    res.scrollIntoView({ behavior: "smooth", block: "start" });
  }

  $("calc").addEventListener("click", calculate);
  $("amps").addEventListener("keydown", e => { if (e.key === "Enter") calculate(); });
})();
</script>
</body>
</html>
```